

Replication Plan

Chiral Spin Order in Kondo–Heisenberg Systems

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Paper Overview

This paper investigates chiral spin order in Kondo–Heisenberg systems on triangular and kagome lattice geometries. The authors study the competition between RKKY-type indirect exchange and Kondo screening in geometrically frustrated lattices, focusing on the emergence of scalar spin chirality and its interplay with charge transport (anomalous Hall effect). The work employs mean-field decoupling of the Kondo lattice Hamiltonian, self-consistent numerical solutions, and variational approaches to map out phase diagrams as functions of Kondo coupling J_K , Heisenberg exchange J_H , and electron filling.

Detailed Workflow

Phase 1: Hamiltonian Construction

1.1 Define the Kondo–Heisenberg Hamiltonian.

- Tight-binding hopping term $H_t = -t \sum_{\langle i,j \rangle} c_{i\sigma}^\dagger c_{j\sigma}$ on the triangular/kagome lattice.
- Heisenberg exchange $H_H = J_H \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j$ between localized spins.
- Kondo coupling $H_K = J_K \sum_i \mathbf{s}_i \cdot \mathbf{S}_i$ between itinerant electron spin and localized spin.

1.2 Construct lattice geometry.

- Build adjacency matrices for 2D triangular and kagome lattices with periodic boundary conditions.
- Support system sizes $N = L \times L \times n_{\text{sub}}$ with $n_{\text{sub}} = 1$ (triangular) or 3 (kagome).

Phase 2: Mean-Field Decoupling

2.1 Implement mean-field decomposition.

- Decouple the Kondo interaction via mean-field parameters: hybridization amplitude $V_i = \langle c_{i\sigma}^\dagger f_{i\sigma} \rangle$ and bond order parameters.
- Introduce chiral order parameter $\chi_{ijk} = \mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k)$ for triangular plaquettes.

2.2 Derive the mean-field Hamiltonian.

- Express the effective quadratic Hamiltonian in momentum space via Fourier transform.
- Diagonalize $H_{\text{MF}}(\mathbf{k})$ for each $\mathbf{k}$ -point in the Brillouin zone.

Phase 3: Self-Consistent Solution

3.1 Implement self-consistency loop.

- Initialize mean-field parameters (hybridization, magnetization, chirality).
- Diagonalize the Hamiltonian at each $\mathbf{k}$ -point.
- Compute ground-state expectation values of all order parameters.
- Update parameters and iterate until convergence ($\Delta < 10^{-6}$).

3.2 Scan parameter space.

- Vary J_K/t and J_H/t over specified ranges.
- Vary electron filling n_c .
- Record converged order parameters at each point.

Phase 4: Phase Diagram Construction

4.1 Identify phases.

- Kondo singlet phase: $V \neq 0$, $\chi = 0$.
- Chiral spin liquid/order: $\chi \neq 0$.
- Magnetically ordered phases: finite sublattice magnetization.

4.2 Map phase boundaries as functions of J_K/t , J_H/t , and filling.

4.3 Reproduce phase diagrams (Figures in the paper).

Phase 5: Transport and Response Functions

5.1 Compute anomalous Hall conductivity σ_{xy} via the Kubo formula or Berry curvature integration.

5.2 Calculate density of states and spectral functions.

5.3 Reproduce transport plots from the paper.

Phase 6: Validation

6.1 Compare phase boundaries with paper's figures.

6.2 Compare numerical values of order parameters at specific parameter points.

6.3 Verify limiting cases (e.g., $J_K = 0$ reduces to pure Heisenberg model).

Datasets

This is a purely theoretical/computational study. No external experimental datasets are required.

Dataset / Source	Type	Location / Notes
Lattice geometries (triangular, kagome)	Synthetic	Generated programmatically from lattice vectors
Model parameters (J_K , J_H , t , filling)	Synthetic	Specified in paper text and figure captions
Paper figures (for validation)	Reference	Extracted from the PDF for quantitative comparison

Models, Tools, and Software

Programming Languages

- **Python 3.8+** or **Julia 1.6+** — for implementing the mean-field solver.
- **Fortran/C++** — optional, for performance-critical diagonalization loops.

Libraries and Frameworks

Tool / Library	Version	Purpose / URL
NumPy	≥ 1.21	Array operations, linear algebra
SciPy	≥ 1.7	Sparse eigensolvers, optimization for self-consistency
Matplotlib	≥ 3.4	Phase diagram visualization, contour plots
LAPACK/BLAS	system	Backend for matrix diagonalization
PythTB	latest	Tight-binding model construction and Berry curvature https://www.physics.rutgers.edu/pythtb/
kwant	≥ 1.4	Optional: quantum transport calculations https://kwant-project.org/

Hardware Requirements

- Standard workstation; the mean-field calculation is per-**k**-point and parallelizable.
- Dense **k**-mesh (e.g., 100×100) requires moderate memory ($\sim 4\text{--}8$ GB).